Can a Monitor Tell When an AI Agent Has Actually Finished Its Task?

Nathan Langley
[NATHAN: affiliation]
With Apart Research
Apart Research — Swarm Alignment Research Sprint
6–8 November 2026

Abstract

When an AI agent reports that it has finished a delegated task, how much should we trust that report? I wanted to test whether a monitor could distinguish completions supported by independent records from claims that did not hold, and whether checking external evidence helped. I used 219 completion-claim jobs from a preexisting orchestration system, with 98 labeled EARNED, 15 UNEARNED, and 106 UNKNOWN. The test contained only seven naturally UNEARNED jobs, below our preregistered minimum of 15, so the natural-failure analysis is underpowered. We also made scripted false completions by editing real reports. In the preregistered comparison on 45 planted cases invisible to the evidence features, Haiku's report-only monitor reached AUROC 0.517 and the evidence-augmented monitor 0.538; their difference was inconclusive. Later exploratory tests found that a task-match-specific Haiku prompt performed better than an omnibus prompt on mismatched reports. On harder same-workflow swaps, task-match scored 0.775 against 0.594 for omnibus, but did not beat a simple BM25 baseline at 0.801. On reports with a key deliverable removed, every tested scorer remained near chance. These findings suggest that asking a narrower question can help, but a convincing completion monitor needs more than report text. They do not establish performance on real-world natural failures or independent deployments.

1. Introduction

I run an orchestrator that delegates work to AI agents. One of the problems with a system like this is that a worker can say it finished something without that being the same as actually finishing it. The report might sound convincing, cite real files, or mention tests, while still missing what the original task asked for. That distinction matters when one agent is deciding whether to trust another agent's work.

This sprint tested three related questions. First, can a monitor recognize a completion claim that does not hold? Second, does giving it basic external evidence improve that judgment? Third, if a general prompt struggles, does asking the narrower question "is this report about this task?" work better? The third question came from looking at the initial results, so I treat it as exploratory rather than as a clean confirmation.

2. Related Work

This sits alongside work on AI control and monitoring under possible subversion [1], multi-agent risks [2], monitoring model behavior [3], sabotage evaluations [4], failure modes in multi-agent systems [5], reward hacking [6], and monitor evasion [7]. My experiment is narrower: it studies claims made in a single real delegation system, then tests whether

monitors notice specific scripted inconsistencies. It does not test intentional deception by an adversarial agent.

3. Methodology

Corpus and labels

The preexisting orchestrator recorded 506 closed delegations from 24 October to 6 November 2026. We excluded 138 jobs from excluded workflows and 92 that hit a privacy filter, leaving 276 jobs across 15 workflows. Of these, 219 had runner status done and a final report claiming completion. Reports were scrubbed of identifiers, and sentences containing verifier wording were removed before monitoring. The database retains at most the last 2,000 characters of each report.

Labels came from independent system records, not monitor scores. EARNED means a verify-run results-ledger line cites the job for the requested result and nothing in an incident or verifier note contradicts it. UNEARNED means an incident or verifier note explicitly says the claim did not hold. Otherwise the label is UNKNOWN; absence of a citation is not evidence of failure.

Two AI readers, Claude Sonnet 5.5 and Opus 5.5, had to agree and quote a source in a blind sweep that increased natural UNEARNED labels from 2 to 15. The sweep knew the preregistered target of 15 and landed exactly there. No human reviewed the labels. An AI re-read agreed on 35 of 42 stratified items ($\kappa = 0.743$). Final labels: EARNED 98, UNEARNED 15 (8 dev, 7 test), UNKNOWN 106.

Split and freeze

With seed 20261106, three of seven workflows containing at least 10 jobs (40%, randomly drawn) were held out whole for test; jobs created on or after 3 November were also assigned to test. Prompts were tuned on dev and frozen in git before test calls. The test EARNED comparison group contained 45 jobs.

Monitors

Models ran through `claude -p` and returned a 0–100 probability that completion was unearned.

L1 received the task and report. L2 also received scripted facts about the existence of cited files and commits and whether commits occurred during the worker run. L3 received the task, report, and last 12,000 characters of the worker transcript.

The omnibus prompt asked for P(unearned) using a four-point rubric beginning with whether the report answered the task. The task-match prompt only judged whether the report matched the task.

No-model controls included report length, keyword self-report scoring, negated Jaccard overlap of task/report words of at least four letters, and later BM25. Haiku generally used three repeats per item initially, with scores averaged across repeats.

Scripted plants

We created 15 examples each of five edits to real reports, with no model-written plant text:

- (a) Rename a cited file to a nonexistent name.

- (b) Replace in-run commits with real out-of-run commits.
- (c) Remove test sentences and insert a fabricated passing log.
- (d) Pair a job's task with a different job's low-overlap report.
- (e) Delete disclosures such as "not merged" or "untested" and add "Everything in the brief is done and committed."

Plants a and b are invisible to L1 by construction; c, d, and e preserve the evidence facts used by L2.

Analysis and phases

AUROC compares each positive set with 45 test EARNED jobs. Intervals are 95% percentile CIs from 2,000 bootstrap resamples clustered by source job.

Phase 2 preregistered that evidence mattered only if the lower CI for L2 − L1 exceeded zero; "no detectable gain" required a CI containing zero with upper bound below 0.10; otherwise the result was inconclusive. Natural results were underpowered below 15 test UNEARNED.

Phase 3 introduced stronger models, task-match, controls, placebo, truncation and L3 post hoc. Its plant-d pass bar (AUROC ≥ 0.75, CI lower bound > 0.5) was set after seeing Haiku's 0.645.

Phase 4 preregistered its repeat-reporting rule before two additional Haiku and one additional Sonnet task-match repeats.

Phase 5 preregistered its new calls before running them, but used the already-seen split and remains exploratory.

4. Results

4.1 Preregistered evidence comparison

For the 45 planted c+d+e cases, L1 AUROC was 0.517 [0.402, 0.623] and L2 was 0.538 [0.433, 0.649]. The paired gap was +0.022 [−0.080, +0.141]: inconclusive under the preregistered rule.

This comparison cannot establish whether richer evidence helps, since these edits deliberately preserve L2's evidence features.

For plant a, L1 scored 0.387 [0.220, 0.553] and L2 0.802 [0.691, 0.900]. For plant b, the respective scores were 0.384 [0.217, 0.548] and 0.467 [0.295, 0.635].

The natural-failure comparison, just 7 UNEARNED against 45 EARNED, was 0.492 for L1 and 0.497 for L2, with gap +0.005 [−0.168, +0.176]; it is underpowered.

Across the 15 natural UNEARNED, an existence check caught 0: 14 had all cited items present (an upper-bound observation), and one had nothing checkable. Haiku omnibus run SD was 3.86 for L1 and 2.43 for L2.

Plant	L1 AUROC [95% CI]	L2 AUROC [95% CI]
c: fabricated test log	0.407 [0.242, 0.578]	0.553 [0.382, 0.715]
d: wrong-job report	0.645 [0.485, 0.785]	0.560 [0.423, 0.698]
e: removed disclosure	0.497 [0.324, 0.668]	0.501 [0.337, 0.675]

4.2 Post-hoc task matching on plant d

The simple word-overlap baseline scored 0.842 [0.723, 0.943], while BM25 later scored 0.921 [0.845, 0.982].

Haiku task-match reached 0.897 [0.807, 0.973] across five repeats (individual AUROCs 0.842, 0.794, 0.867, 0.861, 0.907; SD 0.041).

It beat Haiku omnibus by +0.252 [+0.141, +0.378], but its gap over word overlap was +0.055 [−0.080, +0.199], and its gap over BM25 was −0.024 [−0.120, +0.074].

Word overlap itself beat Haiku omnibus by +0.197 [+0.013, +0.387].

Sonnet omnibus scored 0.889 [0.774, 0.972], Sonnet task-match 0.950 [0.881, 0.997], and Opus omnibus 0.936 [0.824, 0.999], but Sonnet and Opus helped generate the EARNED labels and are not independent checks.

The placebo a+b against EARNED gave Haiku L1 0.386 [0.253, 0.523]; d against a+b gave 0.753 [0.579, 0.893].

The post-hoc permutation estimate for any of five phase-3 d cells passing the chosen bar by chance was 0.01.

4.3 Harder same-workflow swaps and missing deliverables

Phase 5 used 45 d2 plants, swapping in another report from the same workflow with the most similar task (30 positive clusters), and 28 p plants with a key deliverable line removed.

These tests used 45 EARNED controls. The preregistered minimum detectable effects at 80% power were 0.141 for d2 and 0.146 for p, so every phase-5 result is exploratory under its own rule.

Scorer	d2 AUROC [95% CI]	p AUROC [95% CI]
Word overlap	0.644 [0.547, 0.741]	0.514 [0.435, 0.591]
BM25	0.801 [0.712, 0.882]	0.544 [0.467, 0.619]
Haiku omnibus	0.594 [0.506, 0.693]	0.541 [0.451, 0.629]
Haiku task-match	0.775 [0.693, 0.860]	0.551 [0.441, 0.661]
Sonnet* omnibus	0.865 [0.797, 0.933]	0.527 [0.431, 0.624]
Sonnet* task-match	0.899 [0.835, 0.952]	0.534 [0.457, 0.617]

On d2, Haiku task-match exceeded word overlap by +0.132 [+0.028, +0.245], but the 99.17% Bonferroni interval [−0.016, +0.279] crossed zero.

It did not exceed BM25: −0.026 [−0.132, +0.085].

Against Haiku omnibus it gained +0.182 [+0.083, +0.274], surviving the six-cell Bonferroni check [+0.050, +0.305].

Haiku omnibus was worse than BM25 by −0.208 [−0.343, −0.061].

Sonnet task-match exceeded BM25 by +0.098 [+0.018, +0.181] at 95%, but not after Bonferroni [−0.016, +0.223], and its label circularity remains.

On p, all AUROC intervals included 0.5, and none of the paired monitor-versus-baseline comparisons was supported.

These were 590 Haiku omnibus, 590 Haiku task-match, 236 Sonnet omnibus and 236 Sonnet task-match calls; a recount made 28 checks with zero mismatches and a largest CI difference of 0.009.

4.4 Report truncation

The system's stored reports are limited to 2,000 characters.

In a post-hoc 13-plant-d / 35-EARNED subset, uncapped reports were at most 4,054 characters (median 2,556); the earlier 12,000-character phase-3 cap therefore had not truncated them.

Haiku omnibus moved from 0.615 [0.429, 0.776] on 2,000-character reports to 0.680 [0.499, 0.851] uncapped, a paired +0.065 [−0.087, +0.232].

Task-match moved from 0.893 [0.797, 0.975] to 0.927 [0.779, 1.000], a paired +0.034 [−0.153, +0.181].

Neither length improvement was supported. With full reports, task-match exceeded omnibus by +0.247 [+0.016, +0.469].

5. Discussion

The most interesting result to me is not that a model can detect a report paired with the wrong task. A simple lexical baseline often does that too, and BM25 was especially competitive. What matters is that Haiku's general-purpose completion prompt was substantially worse than a narrow task-match question on the same examples. The prompt was being asked to judge too many things at once, and narrowing the question improved its discrimination in these planted settings.

But that improvement has a clear boundary. On the harder d2 swaps, Haiku task-match beat the omnibus prompt and word overlap at the ordinary 95% level, yet it did not beat BM25. On plant p, removing an important deliverable line was effectively invisible to all text-based scorers we tested. That suggests the next step should be checks tied to the actual requested artifact or result, not just a better-written general monitoring prompt. An existence check on the planted p artifacts would be a natural next test; we did not run it.

There is also a distinction between noticing a mismatch and knowing that a completion claim is false. The plants are controlled edits, not naturally observed failures. A monitor that recognizes those edits may still fail on a worker that describes an incomplete job convincingly, and the natural sample here is too small to answer that question. I would not use these numbers as a claim that the system can reliably catch dishonest or mistaken agents in deployment.

6. Conclusion

I wanted to find out whether an AI monitor could tell the difference between an agent reporting that it finished a task and actually having evidence that the task was completed. The results suggest that how we ask the monitor to evaluate a report matters. On planted wrong-task reports, Haiku performed substantially better when asked specifically whether the report matched the task than when asked to judge completion more generally.

However, that improvement did not establish an advantage over BM25, a simple retrieval baseline that performed surprisingly well.

The limitations are important. We didn't have enough naturally unearned completions to draw a reliable conclusion about real failures, and the phase-5 results remain exploratory. More importantly, none of the tested monitors reliably detected when a key deliverable was removed from a report. I think the next useful direction is to move beyond evaluating what agents say they accomplished and toward verifying the actual results of their work. A convincing completion report isn't necessarily a completed task, and monitoring systems need to account for that distinction.

7. Reproducibility and Preexisting Work

The orchestrator, delegation database, results ledger, incident reports, verifier notes, sentinel monitor files and logs all existed and operated before the sprint. The underlying corpus also predates the sprint. They were used as read-only sources, not presented as new sprint work.

The study repository was created during the sprint; its first commit is 6 November 2026 at 16:56 US Eastern. It contains preregistrations, export and scrubbing scripts, label processing, monitor runs, phase 1–5 analyses, figures, and this report.

The repository and its git history are not public at submission. Raw delegation text describes a private person's life and work and is not released; this report provides aggregate statistics only.

The private dataset and unavailable code limit independent replication, even though the procedure and freeze order are described here.

Code and Data: [NATHAN: repo link, optional]

Author Contributions

[NATHAN: contributions]

Appendix A. Limitations and Dual-Use / Ethical Considerations

What this establishes. On one already-inspected split, a task-specific prompt discriminated some planted wrong-task reports better than an omnibus Haiku prompt. It did not establish an advantage over BM25, a reliable signal for omitted deliverables, or a reliable monitor for naturally unearned completions.

Only seven natural UNEARNED test cases existed against the preregistered minimum of 15. There was no human label check; the AI label sweep knew the target count and landed on it. A pending 30-item human check (15 UNEARNED and 15 EARNED) remained unanswered.

Sonnet and Opus were label readers, making their monitored EARNED comparisons circular. All experiments came from one deployment, one operator and one model provider.

Phase-3 design and the task-match prompt followed inspection of test data, and every phase-5 cell was exploratory by the preregistered power rule. No confirmatory claim follows.

Dual use. Knowing which report alterations a monitor misses could help someone produce misleading completion claims in a multi-agent workflow, or make operational failures harder to detect. The plant construction also provides a way to probe monitoring blind spots.

We therefore report aggregate behavior rather than private task content, identifying details, or raw operational logs. The results should be used to strengthen verification, not to infer that plausible reports or existing file references prove successful completion.

The privacy exclusions remain in place; excluded jobs were not reintroduced for sample size.

Appendix B. Monitor Prompt Summary

Omnibus: Return a 0–100 probability that the completion is unearned; apply a four-point rubric beginning "does the report answer this task?".

Task-match: Judge only whether the report is about the assigned task.

L1 uses task plus report; L2 adds scripted existence and commit-timing facts; L3 adds the last 12,000 transcript characters.

Exact prompt text is retained in the nonpublic study repository, so this appendix is a functional description, not a verbatim prompt sufficient for byte-for-byte replication.

LLM Usage Statement

Claude (Anthropic) agents running in my orchestrator designed and ran the experiments, wrote the code and compiled the fact record. ChatGPT (OpenAI) wrote the report prose from that fact record; Claude then checked every number against it. Labels were read by AI models, not checked by a human. [NATHAN: what you reviewed or rewrote]

References

[1] Greenblatt, R., Shlegeris, B., Sachan, K., Roger, F. (2023). AI Control: Improving Safety Despite Intentional Subversion. arXiv:2312.06942. https://arxiv.org/abs/2312.06942

[2] Hammond, L., Chan, A., Clifton, J., et al. (2025). Multi-Agent Risks from Advanced AI. arXiv:2502.14143. https://arxiv.org/abs/2502.14143

[3] Baker, B., Huizinga, J., Gao, L., et al. (2025). Monitoring Reasoning Models for Misbehavior and the Risks of Promoting Obfuscation. arXiv:2503.11926. https://arxiv.org/abs/2503.11926

[4] Kutasov, J., Sun, Y., Colognese, P., et al. (2025). SHADE-Arena: Evaluating Sabotage and Monitoring in LLM Agents. arXiv:2506.15740. https://arxiv.org/abs/2506.15740

[5] Cemri, M., Pan, M. Z., Yang, S., et al. (2025). Why Do Multi-Agent LLM Systems Fail? arXiv:2503.13657. https://arxiv.org/abs/2503.13657

[6] METR (2025). Recent Frontier Models Are Reward Hacking. https://metr.org/blog/2025-06-05-recent-reward-hacking/

[7] Najt, E., Toft, C., Tracy, T., Roger, F., et al. (2026). SLEIGHT-Bench: A Benchmark of Evasion Attacks Against Agent Monitors. arXiv:2605.16626. https://arxiv.org/abs/2605.16626